

Detecting Pre-Collapse Instability in AI Systems via Admissibility Geometry and Multi-View Consistency

Gabe Hospelhorn
Independent Researcher
inquiries@gabehospelhorn.com

May 11, 2026

Abstract

Current AI governance frameworks rely on output-level metrics and post-hoc audits that detect failure only after it manifests in model behavior. We introduce an admissibility measurement layer that estimates distance to instability using a Lyapunov-like potential Φ across three non-redundant observational frames (representational, functional, perturbational), yielding a geometric state signal Δ_Φ and a dynamical flow signal $d\Phi/dt$. We identify a novel failure mode—*parallax instability*—in which systems remain admissible under aggregate measures while becoming internally inconsistent across distinct frames. A noise-aware, conjunctive detection protocol with calibrated thresholds and dwell conditions is provided. In a constructive simulation the parallax signal precedes separatrix crossing, which precedes performance degradation, with ordering robust to parameter variation. This work positions observability as the missing measurement substrate for AI governance.

For practitioners outside representational-geometry research: most AI governance today watches what models do — their accuracy, their loss, their answers — and audits problems after they manifest. We introduce a way to watch the geometry of a model’s internal representations instead, and to compute a single admissibility potential — effectively a stability score — that estimates how close the system is to becoming unstable. The potential is constructed from three independent views of the model, so disagreement between views becomes its own warning sign — visible before behavior degrades.

Contents

1	Introduction	3
---	--------------	---

2	Background and Motivation	3
2.1	The Measurement Gap in AI Governance	3
2.2	Lyapunov-like Stability and Phase Transitions	3
2.3	Representation Space as the Right Domain	4
3	The Admissibility Framework	4
3.1	The Admissibility Potential	4
3.2	Estimators	4
3.2.1	Diversity $\mathcal{B}$ — Effective Rank	4
3.2.2	Persistence $\mathcal{P}$ — Subspace Alignment	5
3.2.3	Coupling χ — Perturbation Amplification	5
3.2.4	Entropy $\mathcal{E}$ — Contradiction Burden	5
3.3	State and Flow	5
3.4	Noise-Aware Separatrix	6
4	Parallax Instability	6
4.1	The Projection Problem	6
4.2	Parallax Divergence	7
4.3	The Parallax Instability Regime	7
5	Constructive Simulation	7
5.1	Purpose	7
5.2	Setup	7
5.3	Three Failure Modes	8
5.4	FTLE Connection	8
6	Discussion	8
6.1	Monitoring Architecture	8
6.2	Limitations	9
7	Related Work	9
8	Conclusion	9
A	Formal Definitions	10
A.1	Detection Protocol	10
A.2	Parameter Calibration	10
B	Supplementary Artifact	10

1 Introduction

Current AI governance frameworks rely predominantly on output-level metrics—accuracy, loss, reward—and post-hoc audits. These approaches share a common failure mode: they detect problems after they manifest in model behavior, not before.

A system need not be failing to be unstable; it need only become inconsistent across distinct observational views.

We introduce a measurement layer that addresses this gap. Its core object is an admissibility potential Φ , defined over representation-geometry snapshots. We decompose Φ across three non-redundant observational frames, and we read two signals from it: a state signal giving distance to the instability boundary, and a flow signal giving rate of margin change.

We identify a new failure mode—**parallax instability**—in which systems remain admissible in aggregate while becoming internally inconsistent across distinct observational views. This condition is detectable prior to both effective separatrix crossing and observable performance degradation.

The contribution of this work is not a new metric, but a detection protocol that integrates multiple observability signals into a conjunctive early-warning system for instability. This enables governance systems to detect instability before it manifests in deployed behavior.

An interactive supplementary artifact is available at: <https://gabehospelhorn.github.io/constraint-geometry/admissibility-monitor.html>

2 Background and Motivation

2.1 The Measurement Gap in AI Governance

Governance frameworks for AI systems typically operate on output projections: task performance, behavioral evaluation, alignment assessments. These are necessary but structurally incomplete. They cannot detect instability that exists in the geometry of internal representations before it propagates to observable outputs.

2.2 Lyapunov-like Stability and Phase Transitions

Lyapunov stability theory provides a framework for analyzing whether a dynamical system remains near an equilibrium [3]. We do not claim a formal Lyapunov proof under general stochastic dynamics. Instead, we construct a Lyapunov-like monitoring functional Φ over representation space and treat its decline as a pre-failure signal.

Unlike a strict Lyapunov function, Φ is not assumed to decrease monotonically under all admissible dynamics. It is instead treated as a monitoring functional whose decline signals increasing instability risk.

2.3 Representation Space as the Right Domain

We define the system state as a representation-geometry snapshot:

$$x_t = \{Z_t, Y_t, R_t\} \quad (1)$$

where Z_t are embeddings on a fixed probe set, Y_t are outputs on fixed prompts, and R_t are responses under controlled perturbations.

3 The Admissibility Framework

3.1 The Admissibility Potential

$$\Phi(t) = \log \mathcal{B}(Z_t) + \log \mathcal{P}(Z_t, Z_{t-1}) - \log \chi(R_t) - \log \mathcal{E}(Y_t, R_t) \quad (2)$$

$$\Delta_\Phi(t) = \Phi(t) - \tau \quad (3)$$

The first two terms represent structural capacity; the latter two represent attenuation pressure.

The logarithmic form serves two purposes. First, it converts multiplicative contributions into additive geometry, allowing independent structural and destabilizing factors to be interpreted as contributions to a single admissibility potential. Second, it ensures scale comparability across terms with different natural magnitudes. The signs are chosen so that increases in representational capacity and persistence raise admissibility, while increases in coupling amplification and contradiction burden reduce it.

Positive contributions therefore enlarge admissibility margin, while negative contributions contract it toward the instability boundary.

Table 1: Admissibility potential terms

Term	Name	Interpretation
$\mathcal{B}(Z_t)$	Diversity / effective rank	Dimensions carrying representational load
$\mathcal{P}(Z_t, Z_{t-1})$	Persistence	Dominant geometry persisting across checkpoints
$\chi(R_t)$	Coupling spread	Small input changes propagating broadly
$\mathcal{E}(Y_t, R_t)$	Entropy / burden	Contradiction under probe pairs

3.2 Estimators

3.2.1 Diversity $\mathcal{B}$ — Effective Rank

$$\mathcal{B}(t) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (4)$$

where λ_i are eigenvalues of the covariance of Z_t [4].

3.2.2 Persistence $\mathcal{P}$ — Subspace Alignment

$$\mathcal{P}(t) = \frac{1}{k} \|U_t^{(k)\top} U_{t-1}^{(k)}\|_F^2 \quad (5)$$

The preceding two terms ($\mathcal{B}$, $\mathcal{P}$) measure structural capacity — how much representational load the system can carry and how stably it carries it. The following two terms (χ , $\mathcal{E}$) measure attenuation pressure — the forces eroding that capacity.

3.2.3 Coupling χ — Perturbation Amplification

$$\chi(t) = \mathbb{E}_i \left[\frac{\|z_t(p_i + \delta_i) - z_t(p_i)\|}{\|\delta_i\|} \right] \approx \mathbb{E}_i [\|J_t(p_i)\|] \quad (6)$$

This measures the expected operator norm of the input–representation Jacobian [7]. The finite-time Lyapunov proxy is:

$$\lambda_\chi(t) \approx \log \frac{\chi(t)}{\chi(t-1)} \quad (7)$$

The FTLE connection is included to relate the admissibility framework to established notions of local exponential sensitivity in dynamical systems. It is included for interpretive alignment; empirical estimation requires measurement from perturbation responses [8].

3.2.4 Entropy $\mathcal{E}$ — Contradiction Burden

$$\mathcal{E}(t) = 1 + \alpha S_t + \beta C_t \quad (8)$$

where $C_t = \Pr[f_t(p_i) \neq f_t(\tilde{p}_i)]$ is the contradiction rate on paraphrase pairs.

3.3 State and Flow

State: $\Delta_\Phi(t) = \Phi(t) - \tau$ — distance to the admissibility boundary.

Flow:

$$\frac{d\Phi}{dt} = \underbrace{\frac{d}{dt}(\log \mathcal{B} + \log \mathcal{P})}_{\text{structural gain}} - \underbrace{\lambda_\chi(t)}_{\text{expansion}} - \underbrace{\frac{d}{dt} \log \mathcal{E}}_{\text{burden growth}} \quad (9)$$

Under moving-aperture conditions, $d\Phi/dt$ captures not only intrinsic system drift but also misalignment between system updates and shifts in the effective admissibility geometry. Instability arises when change occurs faster than adaptation capacity — the rate condition that parallax divergence is designed to detect before boundary crossing.

The pre-collapse regime is formalized as:

$$\mathcal{R}_{\text{pre}} = \{x_t : \Delta_\Phi(t) > 0, d\Phi/dt(t) < 0\} \quad (10)$$

Table 2: Phase portrait and monitoring regimes

Δ_Φ	$d\Phi/dt$	Regime
> 0	> 0	Stable, improving
> 0	< 0	Stable, drifting
≈ 0	< 0	Pre-collapse — intervention indicated
< 0	any	Collapsed regime

with first-order time-to-separatrix estimate:

$$\hat{t}_{\text{lead}}(t) = \frac{\Delta_\Phi(t)}{-d\Phi/dt(t)}, \quad d\Phi/dt < 0 \quad (11)$$

Consistent with moving-aperture formulations, instability may arise prior to boundary crossing when $d\Phi/dt$ is sufficiently negative relative to adaptation capacity. This corresponds to shimmer-like regimes where systems remain admissible by magnitude but unstable by rate.

3.4 Noise-Aware Separatrix

$$\tau_{\text{eff}} = -c \cdot \sigma_\Phi \quad (12)$$

where σ_Φ is estimated from a stable reference window and $c \in \{1.5, 2\}$ is pre-registered.

4 Parallax Instability

4.1 The Projection Problem

Scalar admissibility alone cannot detect frame-incoherent instability, because distinct observational projections may drift in opposing directions while the aggregate potential remains positive. A system can change in ways no single projection captures. Consider a rotation of representation structure that leaves both rank and persistence unchanged: the scalar admissibility reads stable, yet the geometry has moved. Scalar admissibility is therefore insufficient on its own.

We decompose Φ across three non-redundant frames:

Table 3: Three non-redundant observational frames

Frame	Sub-potential	Captures	Misses
Representation	$\Phi_{\text{rep}} = \log \mathcal{B} + \log \mathcal{P}$	Embedding geometry	Behavioral consistency
Functional	$\Phi_{\text{func}} = -\log \mathcal{E}$	Output consistency	Internal geometry
Perturbational	$\Phi_{\text{pert}} = -\log \chi$	Input sensitivity	Semantic content

4.2 Parallax Divergence

$$\Pi_{\text{norm}}(t) = \frac{\text{Var}\left(\frac{d\Phi_{\text{rep}}}{dt}, \frac{d\Phi_{\text{func}}}{dt}, \frac{d\Phi_{\text{pert}}}{dt}\right)}{\sigma_{\text{baseline}}^2} \quad (13)$$

This normalization expresses divergence as a multiple of baseline variance, enabling comparable thresholds across systems [7]. Variance normalization is used here as the minimal scale-independent estimator of cross-frame divergence; covariance-aware generalizations are left for future work.

Sign coherence:

$$S(t) = \frac{1}{3} \sum_i \mathbf{1}[\text{sign}(\frac{d\Phi_i}{dt}) = \text{mode sign}] \quad (14)$$

Parallax instability is not high variance alone, but variance combined with directional incoherence across frames.

4.3 The Parallax Instability Regime

$$\mathcal{R}_{\text{parallax}} = \{x_t : \Delta_{\Phi}(t) > 0, \Pi_{\text{norm}}(t) > c_{\Pi}, S(t) < S_{\min}\} \quad (15)$$

sustained for k consecutive steps.

Table 4: Conjunctive detection protocol

Condition	Requirement	Role
$\Delta_{\Phi}(t) > 0$	System in admissible region	Distinguishes from margin collapse
$\Pi_{\text{norm}}(t) > c_{\Pi}$	Cross-frame variance elevated	Detects divergence magnitude
$S(t) < S_{\min}$	Sign incoherence	Confirms contradiction
k steps	Dwell condition	Suppresses false positives

5 Constructive Simulation

5.1 Purpose

This simulation is not offered as empirical validation. It demonstrates ordering, not magnitude or domain performance. The claim is ordinal: $t_{\Pi} < t^* < t_{\text{perf}}$.

5.2 Setup

$$\mathcal{B}(t) = B_0 e^{-k_B t}, \quad \mathcal{P}(t) = P_0 e^{-k_P t} \quad (16)$$

$$\chi(t) = \chi_0 e^{k_{\chi} t}, \quad \mathcal{E}(t) = E_0 e^{k_E t} \quad (17)$$

Default: $k_B = 0.006$, $k_P = 0.004$, $k_{\chi} = 0.006$, $k_E = 0.005$, $T = 200$.

Performance is modeled as $M(t) = \sigma(\Delta_{\Phi}(t - \ell)/s)$ with lag $\ell = 20$.

5.3 Three Failure Modes

1. **Parallax instability (earliest):** $\Delta_\Phi > 0$, $\Pi_{\text{norm}} > c_\Pi$, $S(t) < S_{\text{min}}$
2. **Trajectory collapse:** $\Delta_\Phi \approx 0$, $d\Phi/dt < 0$
3. **Margin collapse (latest):** $\Delta_\Phi < -c\sigma_\Phi$ for k steps

The observed ordering: $t_\Pi < t^* < t_{\text{perf}}$.

5.4 FTLE Connection

Because $\chi(t)$ is parametrically specified, the FTLE is analytically implied. It is included for interpretive alignment; empirical estimation requires measurement from perturbation responses [8].

An interactive implementation is available as supplementary material (see Appendix B).

6 Discussion

6.1 Monitoring Architecture

The framework defines a four-layer monitoring architecture. These layers operate jointly as a conjunctive detection system, not independent metrics.

Table 5: Four-layer monitoring architecture

Layer	Object	Question
Geometry	$\Delta_\Phi(t)$	Where is the system relative to the boundary?
Dynamics	$d\Phi/dt$	Moving toward or away?
Coherence	$\Pi_{\text{norm}}(t), S(t)$	Do frames agree?
Robustness	σ_Φ, c, k	Is the signal real?

The framework can be situated within a broader failure taxonomy: (1) *capacity failure*, where representational capacity is exhausted; (2) *temporal misalignment*, where system updates lag underlying state change; (3) *model invalidity*, where the assumed geometry is incorrect; and (4) *regulatory failure*, where coordination between system components breaks down. Parallax instability occupies a distinct position as a pre-collapse regime characterized by cross-frame incoherence despite admissibility in aggregate.

Parallax instability can be interpreted as a multi-frame analog of shimmer: instability that arises prior to collapse when system components diverge faster than coordination can maintain coherence. Where shimmer is characterized by rate and anisotropy within a single frame, parallax instability is characterized by directional incoherence across structurally distinct frames.

6.2 Limitations

The simulation demonstrates ordering under specified dynamics but does not validate the framework on real model checkpoints. We treat real-checkpoint validation as the next step, not as a prerequisite for the measurement layer itself. The measurement layer is intentionally decoupled from training dynamics, enabling deployment as post-hoc instrumentation. This work addresses detection, not intervention; control mechanisms operate downstream [1]. The detection layer is substrate-agnostic, but the mechanisms required to stabilize detected instability — prediction, buffering, or abstraction — are substrate-dependent, as formalized in subsequent moving-aperture analyses.

7 Related Work

Loss and accuracy metrics as stability proxies are the dominant approach in ML monitoring [1, 2]. They are output projections and cannot detect structural instability before it manifests behaviorally.

Raju [7] introduces geometric stability as a dimension of representational quality distinct from similarity, measured via split-half correlation of representational dissimilarity matrices. Our persistence estimator $\mathcal{P}$ and coupling estimator χ capture related properties—subspace alignment and perturbation amplification—but embed them within a scalar admissibility potential rather than standalone metrics. Raju [8] identifies a progressive geometric tax under discrete tokenization, consistent with the collapse dynamics our framework is designed to detect.

Effective rank and participation ratio as measures of representational capacity are established in the neural collapse literature [4, 5]. Our use of subspace persistence extends this to temporal stability.

Multi-view consistency as a failure detection principle connects this work to sensor fusion [6]. To our knowledge, no prior work defines a multi-view consistency condition as a distinct instability regime.

8 Conclusion

We have introduced an admissibility measurement layer for AI systems that detects structural instability before it manifests in observable outputs. Core contributions:

- A Lyapunov-like admissibility potential Φ with operationalized estimators
- A two-layer monitoring system separating geometric position from dynamical flow
- A novel failure mode—parallax instability—defined as admissibility under aggregate measure with cross-frame incoherence
- A noise-aware, conjunctive detection protocol
- A constructive simulation demonstrating $t_{\Pi} < t^* < t_{\text{perf}}$ robust to parameter variation

This work separates detection from control, enabling independent evolution of monitoring and governance mechanisms.

A system need not be failing to be unstable; it need only become inconsistent across distinct observational views.

A Formal Definitions

A.1 Detection Protocol

Margin collapse: $\Delta_{\Phi}(t) < -c\sigma_{\Phi}$ for k consecutive steps.

Pre-collapse: $\Delta_{\Phi}(t) > 0$ and $d\Phi/dt < 0$.

Parallax instability: $\Delta_{\Phi}(t) > 0$, $\Pi_{\text{norm}}(t) > c_{\Pi}$, $S(t) < S_{\min}$ for k steps.

A.2 Parameter Calibration

All thresholds calibrated from a pre-collapse reference window T_{stable} :

- $\tau_{\text{eff}} = -c \cdot \sigma_{\Phi}$ from stable window
- $\sigma_{\text{baseline}}^2 = \text{Var}(\Pi(t))$ over stable window
- α (EMA) pre-registered; robustness shown across $\alpha \in [0.05, 0.40]$

B Supplementary Artifact

An interactive reference implementation is available at:

https:
[//gabehospelhorn.github.io/constraint-geometry/admissibility-monitor.html](https://gabehospelhorn.github.io/constraint-geometry/admissibility-monitor.html)

The artifact implements parametric and measured proxy modes and exposes all parameters. All claims are fully reproducible without interaction with the artifact.

References

- [1] Shumailov, I., et al. (2024). AI models collapse when trained on recursively generated data. *Nature*, 631, 755–759.
- [2] Kumar, A., et al. (2022). Fine-tuning can distort pretrained features and underperform out-of-distribution. *ICLR 2022*.
- [3] Lyapunov, A. M. (1907). Problème général de la stabilité du mouvement. *Annales de la Faculté des Sciences de Toulouse*, 9, 203–474.
- [4] Roy, O., & Vetterli, M. (2007). The effective rank: A measure of effective dimensionality. *EUSIPCO 2007*.

- [5] Papayan, V., Han, X. Y., & Donoho, D. L. (2020). Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS*, 117(40), 24652–24663.
- [6] Durrant-Whyte, H., & Bailey, T. (2006). Simultaneous localization and mapping: Part I. *IEEE Robotics & Automation Magazine*, 13(2), 99–110.
- [7] Raju, P. C. (2026). Geometric Stability: The Missing Axis of Representations. *arXiv:2601.09173*.
- [8] Raju, P. C. (2026). The Geometric Alignment Tax: Tokenization vs. Continuous Geometry in Scientific Foundation Models. *arXiv:2604.04155*.
- [9] Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1, 389–399.
- [10] Raginsky, M., Rakhlin, A., & Telgarsky, M. (2017). Non-convex learning via stochastic gradient Langevin dynamics. *COLT 2017*.